What ROI means for AI (important mindset)

ROI for AI is **not just cost savings**. It usually combines:

1. **Hard benefits** (easy to quantify)
    - Cost reduction
    - Productivity/time savings
    - Error reduction
    - Faster cycle time
2. **Soft / strategic benefits** (still monetized, but with assumptions)
    - Better decisions
    - Risk reduction (compliance, quality)
    - Scalability without hiring
    - Employee satisfaction / retention

Standard ROI framework for AI solutions

6-step structure

Step 1: Define the business problem (baseline)

- What process are we improving?
- Who is involved?
- What is the current cost, time, error rate, or delay?

Example
Medical Information team manually responds to HCP queries.
Average handling time = **45 minutes per query**
Volume = **40,000 queries/year**

Step 2: Define the AI solution & scope

Clarify what AI will and will not do.

Example

- AI-powered Medical Q&A assistant
- Handles **Level 1 & Level 2 queries**
- Human review remains for complex cases
- Integrated with approved content repository (RAG-based)

Step 3: Identify costs (Total Cost of Ownership)

Break costs into **one-time** and **recurring**.

A. One-time costs

Cost Item	Example
Data preparation	$60,000
Model development / integration	$120,000
Validation & compliance	$40,000
Change management & training	$30,000
Total one-time	**$250,000**

B. Annual recurring costs

Cost Item	Example
Cloud & inference	$70,000
Support & monitoring	$40,000
Model updates & governance	$30,000
Total recurring/year	**$140,000**

Step 4: Quantify benefits (this is the core)

A. Productivity savings (primary benefit)

Before AI

- 45 min per query
- 40,000 queries/year
- Total effort = 30,000 hours/year

After AI

- AI handles 60% of queries
- Human handling reduced to 15 minutes for remaining
- New effort = 12,000 hours/year

Hours saved = **18,000 hours/year**

If average fully loaded cost = **$70/hour**

💰 **Annual savings = $1.26M**

B. Quality & error reduction (secondary benefit)

- Error/rework rate reduced from 8% → 2%
- Avoided rework + compliance risk

Estimated savings = **$150,000/year**

C. Scalability benefit (strategic)

- Business growth expected: +20% queries/year
- Without AI → hire 5 additional FTEs
- With AI → no additional hires

Avoided cost = **$500,000/year**

Step 5: Calculate ROI metrics

Total annual benefits

Benefit Type	Value
Productivity	$1,260,000
Quality	$150,000
Scalability	$500,000
Total	**$1,910,000**

Net annual benefit

Net Benefit = Total Benefits – Annual Costs
Net Benefit = $1.91M – $0.14M = $1.77M

Step 6: ROI, Payback & NPV

ROI (%)

ROI = (Net Benefit – One-time Cost) / One-time Cost × 100
ROI = ($1.77M – $0.25M) / $0.25M × 100
ROI ≈ 608%

Payback period

Payback = One-time cost / Net annual benefit
Payback ≈ 1.7 months